

VERSION HISTORY

V1	11/02/2025	Internal development
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PROTOCOL

This document contains the outcome specific elements necessary to implement the [post-covid-events-ehrgl protocol](#).

STUDY POPULATION

No additional criteria.

OUTCOMES

Subcategory	Event	Codelists
Arterial thrombosis events	Acute myocardial infarction	SNOMED ICD-10
Arterial thrombosis events	Ischaemic stroke	SNOMED ICD-10
Venous thrombo-embolism events	Pulmonary embolism	SNOMED ICD-10
Venous thrombo-embolism events	Deep vein thrombosis (including during pregnancy)	SNOMED SNOMED ICD-10 ICD-10
Other cardiovascular events	Transient ischaemic attack	SNOMED ICD-10
Other cardiovascular events	Subarachnoid haemorrhage and haemorrhagic stroke	SNOMED ICD-10
Other cardiovascular events	Heart failure	SNOMED ICD-10
Other cardiovascular events	Angina	SNOMED ICD-10
Arterial thrombosis events	Arterial thrombosis events (i.e., any of acute myocardial infarction, ischaemic stroke, and other arterial embolism)	AMI: SNOMED Ischaemic stroke: SNOMED Other arterial embolism: SNOMED AMI: ICD-10 Ischaemic stroke: ICD-10 Other arterial embolism: ICD-10
Venous thromboembolism events	Venous thromboembolism events (i.e., any of portal vein thrombosis, deep vein thrombosis (including during pregnancy), intracranial venous thrombosis, other deep vein thrombosis, and pulmonary embolism)	PE: SNOMED DVT: SNOMED DVT pregnancy: SNOMED Other DVT: SNOMED ICVT: SNOMED Portal vein thrombosis: SNOMED PE: ICD-10 DVT: ICD-10 DVT pregnancy: ICD-10 Other DVT: ICD-10 ICVT: ICD-10 ICVT pregnancy: ICD-10 Portal vein thrombosis: ICD-10

POTENTIAL CONFOUNDERS

We will consider the following potential confounders, which will be defined using the most recent data prior to the study start date:

Confounder	Type	Definition	Data sources
All stroke	Binary	1 if diagnosis present; 0 otherwise	Primary care, HES APC
Other arterial embolism	Binary	1 if diagnosis present; 0 otherwise	Primary care, HES APC
Venous thromboembolism events	Binary	1 if diagnosis present; 0 otherwise	Primary care, HES APC
Heart failure	Binary	1 if diagnosis present; 0 otherwise	Primary care, HES APC
Angina	Binary	1 if diagnosis present; 0 otherwise	Primary care, HES APC
Lipid lowering medications	Binary	1 if prescription present; 0 otherwise	Primary care
Antiplatelet medications	Binary	1 if prescription present; 0 otherwise	Primary care
Anticoagulation medications	Binary	1 if prescription present; 0 otherwise	Primary care
Combined oral contraceptive pill	Binary	1 if prescription present; 0 otherwise	Primary care
Hormone replacement therapy	Binary	1 if prescription present; 0 otherwise	Primary care

SUBGROUPS

Subgroups will be run for the composite outcomes ‘arterial thrombosis events’ and ‘venous thromboembolism events’ only.

PROPOSED OUTPUTS

The proposed outputs from this protocol will be like those included in the paper ‘Association of COVID-19 with arterial and venous vascular diseases: a population-wide cohort study of 48 million adults in England and Wales’. Listed here for convenience:

Table 1. Patient characteristics in the pre-vaccination, vaccinated and unvaccinated cohorts

Table 2. Number of cardiovascular events in the pre-vaccination cohort, vaccinated cohort and unvaccinated cohort, with person-years of follow-up, by COVID-19 severity

Table 3. Adjusted hazard ratios (95% CI) comparing the incidence of arterial thrombotic events after versus before or without a COVID-19 diagnosis, in the pre-vaccination, vaccinated and unvaccinated cohorts, overall and according to COVID-19 severity

Table 4. Adjusted hazard ratios (95% CI) comparing the incidence of venous thrombotic events after versus before or without a COVID-19 diagnosis, in the pre-vaccination, vaccinated and unvaccinated cohorts, overall and according to COVID-19 severity

Table 5. Maximally adjusted hazard ratios (95% CI) comparing the incidence of other vascular events after versus before or without a COVID-19 diagnosis, in the pre-vaccination, vaccinated and unvaccinated cohorts, overall and according to COVID-19 severity

Figure 1. Estimated number of COVID-19 cases identified by community testing in England between January 2020 and January 2022.

Figure 2. Maximally adjusted hazard ratios and 95% CIs comparing the incidence of arterial thrombotic and venous thrombotic events after versus before or without a COVID-19 diagnosis, in the pre-vaccination, vaccinated and unvaccinated cohorts, overall and by COVID-19 severity.

Figure 3. Maximally adjusted hazard ratios and 95% CIs comparing the incidence of arterial thrombotic, venous thrombotic, and other vascular events after versus before or without a COVID-19 diagnosis, in the pre-vaccination, vaccinated and unvaccinated cohorts.

Figure 4. Estimated absolute increase in the risk of arterial thrombotic and venous thrombotic events over time since COVID-19 diagnosis, compared with no COVID-19 diagnosis, in the pre-vaccination, vaccinated and unvaccinated cohorts.